

Long-term Research Directions in Wired Networking

Report of the NSF Community Workshop

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1. Executive Summary

Computer networks are among the most critical infrastructures our society has built. Over several decades, the networking community has identified and resolved a long sequence of important research problems, and has translated many of those results into practice.

The next decade will be different. Machine learning has become the dominant consumer of datacenter capacity, and its communication patterns look very different from the workloads our protocols were designed for. Hardware is evolving rapidly—terabit network fabrics, cache-coherent host interconnects, and specialized accelerators within each host are dissolving the host boundary as a unit of design. And the public Internet is now being used for applications whose requirements mismatch what the Internet architecture promises—all while society's dependence on network infrastructure makes disruption more costly.

This workshop brought together roughly forty participants from computer networking, computer architecture, operating systems, distributed systems, programming languages, security, and theory to identify the long-term research directions that follow. It was organized in two tiers: subarea groups met before the workshop to converge on the questions worth asking within their subcommunities, and the workshop then devoted most of its time to cross-area discussion among all attendees.

Five grand challenges—each tied directly or indirectly to a societal need or to a technology trend—emerged from the workshop.

GC1. Building one machine out of many: a communication architecture for AI datacenters. Machine learning systems now underpin services people increasingly depend on. The ability of these systems, as well as the training and inference cost, is set less by the design of a single accelerator than by the thousands of them distributed across the entire cluster. Training requires those processors to behave as one machine. Advances in technology have ensured that the bandwidth—at least for now—is not the obstacle; instead, the obstacle is the lack of right abstractions that make a heterogeneous collection of processors and links look like one substrate. Accelerators within a host are joined by coherent interconnects, while accelerators across hosts are not; today, this semantic gap—along with the latency and bandwidth gap—is exposed to the layers above. We need new abstractions that hide intra-host and inter-host heterogeneity, hide the coherent and non-coherent divide, and hide failures rather than propagating them upward. We also need new algorithms aimed at end-to-end application-level performance, which here means absolute tail communication latency rather than average; and new algorithms for multipath and traffic engineering suited to fabrics whose endpoints are accelerators. Machine learning algorithms and hardware are both emerging rapidly, making this one of the most pressing grand challenges facing our community.

GC2. An Internet that survives: resilience for infrastructure we now depend on.

The Internet is one of society's most critical infrastructures. Unfortunately, as our reliance on the Internet grows, disruptive events have also become more frequent and larger. The assumption that the so-far robust architecture will hold is increasingly hard to defend. We need new architectural principles for resilience. We need to reassess threats, endogenous and exogenous. We need network verification beyond the hyperscalers. And we must pursue all of this in concert with closely interconnected critical systems such as the power grid. Dependence on the Internet is deepening faster than our understanding of how it fails, leaving little time to design resilience rather than retrofit it later.

GC3. Application-defined networking without losing generality. Emerging applications running atop the computer networking infrastructure (e.g., AR and VR) place higher demand and stricter requirements on the last mile, and last-mile infrastructure has not kept up. The core obstacle is the application-neutral interface, a fundamental principle around which existing computer network infrastructure and protocols were designed. We need new abstractions that let applications control their own traffic, perhaps by enabling the network to publish resource-control requirements while an application control plane performs application-driven allocation. We also need to revisit the infrastructure abstractions that are most useful to applications, e.g., low-level primitives such as virtual machines and containers, or high-level services such as databases and collective compute, and whether accelerators should be exposed or abstracted over. We need to revisit protocol design with applications in mind, e.g., offering partial rather than perfect reliability given that many modern applications (e.g., IoT, AR, VR, and robotics) need bounded latency more than perfect reliability. The obstacle is not the shared operating system or the network stack; it is hard-coded policies inside them, often fixed independent of the application they serve. Rearchitecting the entire stack—from hardware to protocols to operating systems stack—from the perspective of application needs is one of the most critical questions for our community today.

GC4. Robust network designs: new theoretical foundations. The networking infrastructure has evolved rapidly over the past 15 years. Host interconnects have evolved to include multi-terabit links and switches and are increasingly heterogeneous in terms of bandwidth, functionality, and semantics. Datacenter network fabrics have evolved in architecture (a few hyperscale datacenters to multiple microdatacenters to many neo datacenters), in technologies (copper to optical to reconfigurable), in topologies, and in switch functionalities. The protocols—from link layer to network layer to transport layer—have also evolved accordingly. Unfortunately, this evolution has outpaced the development of theoretical foundations for reasoning about the networking infrastructure. The community needs new ones. We need new theoretical foundations for traffic engineering for datacenters with directly-networked accelerators and for inter-datacenter traffic; for routing and topology design in reconfigurable networks; for multipath transports now that endpoints perform some of it; for flow control within host networks; and for understanding whether predictable machine learning workloads can be planned for in advance. These foundations must also connect to GC5, so that robustness established in a model holds in deployment. Reconfigurable fabrics, accelerator-dense datacenters, multipath transports, and new host networks are being built and operated today based on intuitions that are not backed by theoretical foundations, which makes closing the gap pressing.

GC5. Closing the deployment gap: evaluating network mechanisms. Society depends more than ever on infrastructure it rarely sees—the public Internet, datacenters, and the services built on both. Yet the performance a mechanism shows in a research paper and the performance it shows in deployment often differ widely. We need mechanisms to bridge this gap. Two complementary approaches exist: the empirical, which extends what we can observe in simulation and experiment, and the theoretical, which establishes performance bounds. Both must begin with more thoughtful assumptions—about topologies, failure modes, and how faithfully a decentralized, partially observable, sometimes adversarially changing, heavy-tailed setting can be modeled at all—and with agreement on what to measure: meaningful

performance metrics and the workloads a design is evaluated against. We need to build publicly available, verifiable measurements collected over time—perhaps by converging on a shared tool and establishing a path to sustainability that does not depend on a company's commercial interest. We also need new theoretical models and tools that enable modeling the infrastructure with the above assumptions and metrics. Infrastructure is being built faster than we can evaluate and understand it, making it critical to reduce the gap between research and deployment.

2. Introduction

The computer networking infrastructure has evolved rapidly over the last decade to accommodate cloud computing and distributed analytics; the coming decade promises changes at least as large.

The workshop set out to answer a deliberately broad question: what are the long-term research directions in wired networking over a roughly decade-long horizon? The workshop was planned to be cross-disciplinary—networking, operating systems, hardware, theory, measurements, and programming languages researchers in the same room.

This report is organized as follows. Section 3 describes the workshop's organization, which is transferable to other community efforts. Section 4 summarizes the framing perspective offered by the opening keynote. Section 5 presents the workshop's technical synthesis. Section 6 concludes.

3. Workshop Organization

A workshop convening seven subareas must find the right balance between depth within each and conversation across them. To achieve this goal, the workshop used a two-tier structure that strikes that balance in time: convergence within a subarea happens before the workshop, leaving the workshop itself for cross-subarea discussion among participants.

3.1 Tier zero: white papers

Ahead of both tiers, the broader community was invited to submit a one-page white paper on their favorite long-term research direction in wired networking. The call was issued through NSF's mailing list, SIGCOMM's list, and social media.

The white papers gave the broader community a channel to inject ideas and long-term research directions into the agenda. Submitting a white paper was not a precondition for invitation, and the final invitation list combined white-paper authors with other researchers, with deliberate effort to recruit interdisciplinary attendees.

3.2 Tier one: pre-workshop subarea discussions

Seven subareas were designated for dedicated discussion, roughly three hours long, in advance of the main workshop:

- Networked applications
- Network measurements
- Network protocols
- Network hardware
- Network software and operating systems
- Network theory
- Network security

Each group was led by recruited subarea leads. At least one of the two co-chairs attended every pre-workshop discussion, giving organizers a global view of how the groups were converging and where they overlapped.

Each group's stated goal was to filter and summarize the ideas circulating in its subarea so the main workshop could focus on cross-subarea discussions. The groups also developed a shared understanding of what was worth addressing within their own subarea, and the interval between the pre-workshop discussion and the workshop gave participants time for protracted, offline follow-up conversations.

Each group's concrete deliverable was a slide deck capturing its ideas, to be used to drive discussion at the workshop. Five of these decks—covering network hardware, programming languages, the public Internet, network and operating systems, and network theory—became the opening material for the corresponding workshop sessions and form the basis of Section 5 of this report.

3.3 Tier two: the main workshop

The workshop itself ran a day and a half at the Verizon Executive Education Center on the Cornell Tech campus in New York City, with a hybrid option for remote participation. It opened with an industry keynote and then moved through six cross-area sessions:

Day	Session
Friday, September 15	Keynote: challenges in network research
Friday, September 15	Pondering the public Internet
Friday, September 15	Network hardware
Friday, September 15	Application-driven networking
Friday, September 15	Panel discussion, over dinner
Saturday, September 16	Network and operating systems
Saturday, September 16	Network theory
Saturday, September 16	Programming languages techniques for networking
Saturday, September 16	Wrap-up

Each session opened from the deck that the subarea group had already converged on; thus, sessions could begin at the point of cross-area discussions. A few graduate students from the workshop organizers' groups attended specifically to capture discussion, and each session had an accompanying shared document for collaborative note-taking during the session.

Attendees were expected to participate in both tiers. NSF program staff attended throughout and gave introductory remarks.

4. Perspectives from Practice

The workshop opened with a keynote from David Maltz, Technical Fellow and Corporate Vice President for Azure Networking, speaking in a personal capacity.

The scale framing was direct: a global network spanning sixty-one regions, more than 175,000 miles of fiber and subsea cable, and more than 20,000 peering connections, carrying on the order of fifteen exabytes per day. The applications riding on it include 911 dispatch, hospital IT, banking, and the movement of goods worldwide. Quality in this infrastructure, he argued, is not an engineering preference but a public safety property.

Five observations stood out from the keynote:

- **AI supercomputing is a networking problem.** The challenge is moving enormous

amounts of data among many endpoints that are specialized hardware ASICs. The gaps named were transport libraries with hardware-friendly implementations, scheduling and congestion control, and a network layer with virtualization capability.

- **Specialized layers are temporary.** The interconnect layers currently separating the "back-end" accelerator network from the "front-end" network are, on this argument, short-lived: the right move is to make the whole thing one network. That raises a coherent research agenda—naming, virtualization and multi-tenancy, what verbs to support, flow control, error control, in-order delivery—for what becomes the essence of the disaggregated datacenter.
- **The physical layer is a live source of disruption.** Hollow core fiber, in which light travels through gas rather than glass, is roughly 47% faster than traditional fiber. This expands the region search area by a factor of 2.25. Co-packaged optics, by contrast, was presented as mostly an engineering and supply-chain problem rather than a physics one—but one that will radically change the design of servers, coprocessors, accelerators, racks, and networks. Interference and alignment are the challenges to resolve before wireless and free-space optics can be used inside the datacenter.
- **Control-plane scale is its own research problem.** Azure has more than 100 million SDN policy objects under management and 2.5 billion policy-object operations per day, with customers expecting to create or destroy 20,000 virtual machines per minute and to address a million IPs per virtual network. The standard remedy—partition the state—collides with customers' expectation of seamless global updates and peering across fault domains. Managing SDN policy, he argued, is now as complicated as managing network devices ever was, and the correctness and consistency of goal state under continuously updated code is an open systems problem.
- **Denial-of-service defense is uneven across layers.** Denial-of-service attacks remain challenging. Large L3-L4 attacks are generally well handled, even at terabit scale, but L7 attacks became more common through 2023 and are harder to defend, which raises the question of whether large models could help generate attack signatures. Beyond volumetric attacks, the harder problem is making secure applications the default outcome of ordinary development: better sandboxes and taller walls are not enough, certificates remain difficult to rotate and verify, and building extensible or updatable services safely remains unsolved, as Log4j and malware in open-source repositories illustrate.

5. Research Challenges and Opportunities

This section synthesizes the technical content of the workshop, organized by area. The subsections run in the order of the grand challenges each area speaks to most directly, though several areas contribute to more than one:

Area	Grand challenges it contributes to
5.1 Network hardware	GC1, GC3, GC5
5.2 Programming languages and verification	GC2, GC3
5.3 The public Internet	GC2, GC3, GC5
5.4 Network and operating systems	GC1, GC3, GC4, GC5
5.5 Network theory	GC1, GC4, GC5

5.1 Network hardware

The discussion started with the observation that hardware innovation is critical to all types of networks, and that while the networking community has made significant strides in software stacks, protocols and applications, its footprint in hardware design has historically been smaller—not for lack of interest. Five themes were identified.

- **Machine learning as a first-class workload.** The network is a bottleneck for ML training, and is increasingly becoming a bottleneck for inference. This calls for software-hardware co-design and specialized coprocessors, transport libraries, congestion control, and virtualization optimized for this workload. The community should pay attention to training, inference, large language models, and mixture-of-experts structures. We need to identify the benefits and feasibility of in-network acceleration for this workload. We, as the networking community, also have a new metric to think about: Energy. For example, training a frontier model consumes millions of GPU hours, serving inference for one consumes millions of kilowatt-hours a month, and the PCIe physical layer carries a disproportionate energy footprint.
- **Emerging hardware.** Co-design opportunities with optics, including novel fiber technologies, co-packaged optics and silicon photonics in multi-chip modules, and analog photonic computing; with CXL, which invites rethinking node and rack architecture around cache-coherent interconnects for devices and not only memory, and around nodes participating in multiple networks; and with novel switch and NIC designs.
- **Integration of hardware with applications.** How does one design and test new hardware devices together with the applications that will use them? Open questions include meaningful performance metrics, which workloads to evaluate, the relative roles of FPGA, P4, emulation, simulation, and ASIC synthesis, how to evaluate at scale, and the programmability, compiler, and abstraction layers in between.
- **Infrastructure and testbeds.** Networking testbeds—CloudLab, FABRIC, Chameleon, the San Diego Supercomputer Center, and others—are vital, but they suffer from unsynchronized hardware and software ecosystems and difficult system integration. The barrier to entry is high. The group asked directly how hardware, prototypes, and testbeds should be shared, noting that what needs sharing is not only the hardware platform but the software stacks, lessons, and accumulated experience, and how a group can realistically start a new hardware-oriented research agenda.

The discussion also asked which applications benefit most from hardware innovation, what testbed the research community actually needs, and whether quantum networking is a hardware problem at all and whether it is relevant on this timeline. There was also quite a bit of discussion on programmable network hardware. However, the workshop attendees did not reach consensus on the future of programmable network hardware. The attendees also noted that several important kinds of network hardware were not discussed: emerging questions on host interconnects, optical interconnects, quantum network hardware, and wireless hardware.

5.2 Programming languages and verification

The discussion identified four directions.

- **Modular reasoning.** Layering gave the field scalability, end-to-end properties, local checking and enforcement, and debuggability. Modern deployments have complicated the picture—service meshes and VXLAN tunnels mean the stack is no longer a clean sequence. The research agenda is abstractions for "domains," formal interfaces between domains, synthesis and verification across them, handling uncertainty and incomplete information, and supporting new functionality within this discipline.
- **Verification for modern control planes.** The control plane is no longer BGP among routers; it is network controllers, DNS, and storage systems with dependencies among them. The October 2021 Facebook outage served as the motivating case: billions of users lost access for hours, and recovery was slow. The agenda is to formalize these dependencies, prove resilience to all k-failure combinations and to disasters, and establish static stability.
- **Network verification beyond the hyperscalers.** Verification works where there is expertise and specification. Elsewhere, it does not reach. Obstacles include a lack of precise specifications, limited user expertise, and the need for new mental models and abstractions. The proposed approach draws inspiration from software testing—

type systems, testing frameworks, programming environments—and is willing to trade strong guarantees for accessibility, to pursue verification without formal specifications, and to treat this as a problem at the intersection of programming languages, networking, and human-computer interaction.

- **An ecosystem of languages for networks.** Well-designed languages enable reuse, composition, fast and safe refactoring, and correct-by-construction design; P4 for switch data planes, NetKAT for network-wide data planes, Propane for network-wide BGP, etc., are existing instances. Unexplored domains include middleboxes, service meshes as application-layer networks, richer control planes, and the enhanced Internet. Beyond individual languages, the goal stated was a thriving ecosystem—a DSL for each domain, shared intermediate representations, and programming environments.

5.3 The public Internet

The discussion converged on three threads: new abstractions for a new Internet, managing resources in good times and bad, and gaining confidence in what we deploy.

- **Application-aware resource management.** Demand on the last mile has risen, and its requirements have tightened, driven by applications such as AR and VR, while last-mile infrastructure has not kept pace. Network resource management is a well-studied problem, but deployed mechanisms lack application awareness—held back by two barriers: a philosophical one that requires the network to be general and neutral, and a pragmatic one that makes it difficult to deploy. The direction identified was to decouple decision-making from enforcement, offering parameterized abstractions that let applications and users control their own traffic, and to limit deployment to the last mile, where it is both easier and most needed. Another proposed architecture has the network publish resource-control requirements, and an application control plane realize globally application-driven allocation. Related open questions include what abstraction the infrastructure should expose so that the application control plane can see how application-level usage maps onto infrastructure resources, and how application-defined allocation should interact with the network's own adaptation.
- **Computation-centric networking.** Cloud providers rent virtual hardware and are paid for effort, and they have little visibility into application dataflow—tasks spend their time waiting for data, but the dataflow is only revealed after placement, during execution. The alternative framing is that the network exists to provide answers to computation, which requires visibility into dataflow, a common and objective notion of correctness, and separation of I/O from compute. The payoff would be ultra-high-density multi-tenancy, heterogeneous infrastructure, massive burst parallelism, and verifiable computation, along with a shift from paying for effort to paying for results.
- **Internet survivability.** The Internet is a critical cyber-physical infrastructure that was not designed as one, and the working assumption that its robust architecture will hold is increasingly hard to justify as both dependence and the frequency of disruptive events grow. A research agenda on resilience was proposed, comprising an understanding of the architectural principles resilience should rest on—one suggestion being to revisit network control system architecture using insights from multi-layer sensorimotor control systems—a reassessment of both endogenous and exogenous threats, and coordination with interconnected critical systems such as the power grid.
- **Gaining confidence: public decision flows.** Most reinforcement-learning algorithms for networking never get deployed. The Internet is a uniquely difficult setting to simulate, and reinforcement learning needs simulation because training on a real network is too slow. The proposal was to proceed in stages: use simulators and emulators as hypothesis generators and a first selection stage, then test a small number of algorithms against a small fraction of real traffic. The unit of sharing would not be datasets or dataflows, but *decision flows*—sequences of opportunities to make decisions affecting representative traffic in randomized fashion, under a quality constraint. The illustrative example was adaptive bitrate selection on 0.01% of a large streaming provider's video traffic, subject to the condition that treated traffic

achieves at least 95% of the quality of experience of untouched traffic.

- **Gaining confidence: verifiable monitoring data.** Evaluating the quality of Internet connectivity requires publicly available, verifiable data collected over time. This is an old problem that the pandemic renewed interest in, but essentially nothing of the kind exists outside private companies; there is no single tool the community agrees on, and the current approach—deploy a device that must be paid for, maintained, and troubleshot—has no sustainable financial model absent a company. Debugging home networks is getting harder as the number of home devices grows.

5.4 Network and operating systems

The discussion was organized around a sequence of questions rather than a list of topics, and closed with two forward-looking directions.

- **What are the right metrics of success?** The group discussed several metrics of success: provider-centric metrics such as throughput and latency SLAs, vCPU oversubscription rate, and VM memory allocation; application-centric metrics such as per-job completion time relative to deadline or importance, and per-application quality of experience; and long-term metrics that are rarely made explicit—the barrier to innovation and entry in cloud computing, and whether a healthy competitive ecosystem of cloud providers still exists in a decade. These metrics operate on different timescales, which is part of what makes them hard to trade off.
- **What are the right abstractions between application and provider?** The group discussed several related open questions: low-level primitives such as VMs, containers, and functions, or high-level services such as databases, collective compute, and model training and inference? Is there a prudent separation between compute and I/O? Should accelerators and specialized hardware be exposed or abstracted over? How should an application convey figure-of-merit metrics to the infrastructure? And, since resource allocation requires predicting the consequences of a decision up front, how does infrastructure acquire that predictive ability?
- **Where should functions live?** User space, kernel space, or hardware—and how does that decision connect to computer architecture, operating systems, distributed systems, storage, and networking as separate research communities?
- **Network design for next-generation applications.** IoT, AR/VR, and robotics impose stringent throughput and latency requirements but may not require perfect reliability, which raises the question of partial reliability, of queue-management schemes that drop semantically less useful packets, of how a private network customized to a digital farm or smart factory should differ from a general one, and of what parameters and abstractions the network must expose to permit that customization.
- **Data-driven networking 2.0.** The first generation of data-driven networking used available data to add network functionality; the group discussed that the second is likely to treat data as the narrow waist—richer features from better monitoring, explicit tradeoffs between what data to collect and what algorithm to design, and consequences for protocol stacks and implementations. The claimed benefits are a rich algorithmic and implementation playground spanning software, hardware, and architecture, informed rather than ad hoc decisions, and freeing developers from incremental heuristic development to focus on architectural questions.

5.5 Network theory

The discussion proposed two tests for when theory belongs in a discussion of future networking research: when an area is established, but its solution bottleneck is a theoretical question, and when the intellectual coherence of a domain is crying out for theory. It named two audiences: the networking community stepping back to see what it needs, and theorists who might be recruited. Four challenges were identified.

- **Traffic engineering and topology design.** Multi-commodity flow is linear and solvable with an LP optimizer, which sounds like a solved problem and is not. Major

clouds have used MCF optimizers for over a decade, and scale has outrun them: a recent result reports 1,500 nodes and 8,000 edges taking 5.5 hours in a commercial solver, which forces approximation for speed. Open directions include faster MCF; traffic engineering for emerging datacenter networks with directly-networked accelerators; joint MCF and topology design for reconfigurable networks, where recent work has established both the value of workload-obliviousness and lower bounds showing that low latency and high utilization are hard to achieve simultaneously; the question of what lies between oblivious routing and centrally optimized MCF in the hierarchy of routing schemes, and what those intermediate points can and cannot do; whether predictable ML workloads can be exploited to plan traffic engineering in advance; and the joint problem combining MCF, topology design, workload placement, and congestion control.

- **A theory of multipath.** Endpoints now perform some of the traffic engineering—MPTCP, MPLS, TeXCP, SD-WAN, and an expanding set of multipath transports spanning wired, 5G, and Wi-Fi paths. The community is doing this without understanding it: how many paths are enough, what happens when endpoint multipath combines with network traffic engineering, and what happens with many end hosts doing it simultaneously. Semi-oblivious routing appears to be an apt model, and its original theoretical results were pessimistic while the technique works well in practice—a gap suggesting some aspect of reality is not being modeled, and one where there has been recent progress.
- **Trustworthy machine learning for systems.** A self-optimizing, self-monitoring, self-repairing, self-securing cloud is an attractive goal, and a new ML-based resource allocator or load balancer that works in testing puts thousands of machines at risk when deployed. The concern is not exclusive to machine learning—heuristics and LP optimizers raise it too—but ML sharpens it, because behavior outside the training regime is uncertain and because the models resist intuitive understanding and theoretical analysis. The proposed direction is to incorporate machine learning as *advice with guarantees*: for instance, that a prediction within ϵ of correct implies a decision within $\epsilon + \delta$ of optimal, or that good advice implies an NP-hard solver runs in polynomial time. A concrete application would be using predicted network properties such as queue lengths to inform congestion control decisions.
- **A modern theory of congestion control.** Traditional theory produced real insight into stability, convergence, utility maximization, and fairness, and it does not describe the algorithms actually deployed. There is a wide gap between control-theoretic or Kelly-style models and modern data-driven protocols that handle unpredictability, multiplicative increase, and diverse utility functions. The question is what the theory of BBR, Copa, and PCC on the Internet, and of Swift, HPCC, NDP, Homa, and dcPIM in the datacenter, actually is—what the right metrics are, whether regret minimization is a useful one, how to model protocol diversity, and what the models are missing given that deployment performance often fails to match prediction.

Discussion at the workshop added two points to this group's agenda: that the community needs to state its questions in a form on which theorists can make progress, and that the scope should extend past optimization to the question of how an Internet ecosystem evolves.

6. Conclusions

The last few years have been among the most exciting in the history of computer networking: the emergence of new host interconnects such as CXL and specialized accelerator interconnects, the rapid evolution and adoption of Terabit network fabrics, and new applications—e.g., machine learning, AR/VR, robotics—that depend directly on the network infrastructure. While exciting, these developments also present new challenges and research directions for the wired networking community. The workloads running atop the infrastructure have shifted fundamentally. Hardware changes both

enable and require new solutions in software stacks and in network protocols. And while our reliance on the network infrastructure has increased, disruptions have become more frequent and larger.

This workshop brought together researchers from computer networking, computer architecture, operating systems, distributed systems, programming languages, and theory to identify the long-term research directions that follow. Over multiple weeks of pre-workshop discussions and one and a half days of in-person discussions, the workshop produced five grand challenges for the community. These challenges are interrelated and require an interdisciplinary approach to make progress.

The interdisciplinary workshop discussion and outcome were in part successful due to the two-tier structure: subarea discussions in smaller groups of experts before the workshop, and synthesis across subareas during an in-person discussion. We believe that this workshop structure is worth carrying forward independent of the workshop findings.

Appendix A: Workshop Participants

Organizers

Name	Affiliation
Rachit Agarwal	Cornell University
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NSF program staff

Name
Darleen Fisher
Joseph Lyles
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Participants

Name	Affiliation	Area
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Fabián Bustamante	Northwestern	Network measurements and applications
Manya Ghobadi	MIT	Network hardware
P. Brighten Godfrey	UIUC	Network hardware, theory
Srikanth Kandula	Microsoft Research	Network security
Ethan Katz-Bassett	Columbia	Network measurements and applications
Samira Khan	University of Virginia / Google	Network hardware
Ming Liu	Wisconsin	Network hardware
Ratul Mahajan	University of Washington	Networking and PL
David Maltz	Microsoft Azure	Network software, keynote
Radhika Mittal	UIUC	Network protocols and applications
Lai Yi Ohlsen	Measurement Lab	Network measurements
Aurojit Panda	NYU	Network software
KK Ramakrishnan	UC Riverside	Network protocols
Vyas Sekar	CMU	Network security

Scott Shenker	UC Berkeley	Network architecture
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David Walker	Princeton	Networking PL and theory
Anduo Wang	Temple	Network protocols
Walter Willinger	NIKSUN	Network security, measurements
Keith Winstein	Stanford	Networked applications
Richard Yang	Yale	Networked applications
Minlan Yu	Harvard	Network software
Pamela Zave	Princeton	Network protocols

Cornell student rapporteurs and support

Name	Role
Saksham Agarwal	Rapporteur
Qizhe Cai	Rapporteur
Midhul Vuppalapati	Rapporteur
Lizette DeJesus	Event coordination

In addition, Antoine Kaufmann, Robert Kleinberg, Todd Millstein, and Vishal Shrivastav also contributed to pre-workshop discussions.

Appendix B: Agenda

Friday, September 15, 2023—Verizon Executive Education Center, Room 225, Cornell Tech

Time	Session
9:30–10:30	Keynote—David Maltz, Microsoft Azure
11:00–12:30	Pondering the public Internet
2:30–4:00	Network hardware
4:30–6:00	Application-driven networking
6:00–8:30	Panel discussion, with dinner

Saturday, September 16, 2023

Time	Session
9:00–10:00	Network and operating systems
10:15–11:15	Network theory
11:30–12:30	Programming languages techniques for networking
12:45–1:00	Wrap-up